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Urban flood nowcasting techniques

Latest Approaches to Urban Flood Nowcasting: Integrating Rainfall Forecasts with Drainage Network Models

Quick Reference

Key Findings Table

Theme/Approach	Description	Key References
Hybrid AI + Physics-based Integration	Deep learning (LSTM, CNN, transformers) fused with hydrodynamic models for rapid, accurate nowcasting	1 2 3
Graph Neural Networks (GNNs)	GNNs model drainage as graphs, capturing spatiotemporal dependencies for improved flood prediction	4 5
Multi-source Data Fusion	Radar, satellite, CML, IoT sensor fusion enhances rainfall forecast resolution and reliability	6 7 8
Surrogate Modeling for Efficiency	Deep neural networks and Gaussian Process emulators replace full hydrodynamics for real-time prediction	9 10 11
Uncertainty Quantification	Ensemble, Bayesian, and Monte Carlo methods propagate rainfall forecast uncertainty through flood models	6 7 8
Explainable AI (XAI)	SHAP, LIME, and attention mechanisms increase transparency and trust in hybrid models	5 12 13

Direct Answer

Recent approaches to urban flood nowcasting integrate high-resolution, multi-source rainfall forecasts (leveraged via advanced AI techniques including deep learning, hybrid ensemble methods, and graph neural networks) with sophisticated urban drainage network models such as coupled 1D-2D hydrodynamic simulations and surrogate models. These frameworks employ data fusion from radar, satellite, microwave links, and IoT sensors, and use ensemble and uncertainty quantification methods (e.g., Bayesian optimization and MPC frameworks) to produce near-real-time predictions with improved accuracy and reduced computational costs.

Study Scope

- **Time Period:** 2018–2024
- **Disciplines:** Hydrology, Urban Engineering, AI/ML, Remote Sensing, Environmental Informatics
- **Methods:** Deep learning, hybrid AI-physics modeling, graph neural networks, surrogate modeling, data fusion, uncertainty quantification, high-resolution hydrodynamic simulation, explainable AI

Assumptions & Limitations

- **Assumptions:**
  - Urban drainage network data and high-resolution rainfall observations are available or can be estimated via remote sensing and sensor networks.
  - AI/ML models are trained on representative historical and/or synthetic data.
- **Limitations:**
  - Data scarcity and quality issues, especially for drainage network details and real-time validation.
  - Computational constraints for large-scale, high-resolution real-time applications.
  - Uncertainty quantification and propagation remain challenging, especially in operational settings.
  - Generalizability across diverse urban morphologies and climates is not fully established.

Suggested Further Research

- Systematic, multi-city evaluation of integrated nowcasting frameworks under diverse meteorological and urban conditions.
- Improved, standardized methods for uncertainty quantification and propagation from rainfall forecasts through to flood impact predictions.
- Development of adaptive, explainable, and scalable hybrid frameworks that integrate human-sensed and physical data streams.
- Exploration of reinforcement learning and continual learning for adaptive flood nowcasting in evolving urban environments.

1. Introduction

Urban Flood Nowcasting: Context and Challenges

Urban flooding is an escalating risk due to rapid urbanization, climate change, and the increasing frequency of extreme rainfall events. Traditional flood forecasting methods, often reliant on sparse rain gauge networks and static drainage models, struggle to provide timely and accurate warnings in complex urban settings. The integration of high-resolution, real-time rainfall forecasts with detailed urban drainage network models is now recognized as essential for effective urban flood nowcasting, enabling proactive flood management and risk mitigation [3] [14] [15].

However, this integration faces significant challenges:

- **Data heterogeneity:** Rainfall and drainage data come from diverse sources with varying spatial and temporal resolutions.
- **Computational demands:** High-fidelity hydrodynamic simulations are computationally intensive, limiting real-time applicability.

- **Uncertainty:** Both rainfall forecasts and drainage models are subject to significant uncertainties, which must be quantified and managed for reliable decision-making.

This report synthesizes the latest research on methodologies that bridge rainfall forecasting and urban drainage modeling, focusing on hybrid AI-physics approaches, graph neural networks, multi-source data fusion, and uncertainty-aware frameworks.

2. Theoretical Frameworks

2.1 Rainfall Forecasting in Urban Environments

AI and Hybrid Models for Rainfall Forecasting

- **Deep Learning Architectures:** LSTM, CNN-LSTM, and transformer-based models capture complex temporal and spatial rainfall patterns, outperforming traditional statistical and machine learning methods [16] [17] [18].
- **Hybrid Models:** Integration of Adaptive Neuro Fuzzy Inference Systems (ANFIS) with metaheuristic optimization (e.g., Artificial Bee Colony, Firefly) further improves accuracy and robustness [19] [20].
- **Ensemble Methods:** Combining outputs from multiple models (e.g., DNNs, NWP) via ensemble learning (e.g., XGBoost, CatBoost) enhances prediction skill and reduces uncertainty [21] [22].

Remote Sensing and Commercial Microwave Link (CML) Techniques

- **Radar and Satellite:** Provide high spatial and temporal resolution rainfall estimates, critical for initializing hydrological models [23] [24] [25].
- **CML:** Cellular network microwave links offer dense, near-ground rainfall monitoring, especially valuable in urban areas lacking traditional sensors [6] [26] [27].
- **Integration:** Calibration and fusion of radar, satellite, and CML data reduce biases and improve rainfall field accuracy [28] [29].

Video-Based and Crowdsourced Rainfall Measurement

- **Video-Based Methods:** Real-time rainfall estimation using surveillance and smartphone cameras, enhanced by deep learning for rain streak detection and intensity estimation [30] [31] [32].
- **Crowdsourcing:** Networks of personal weather stations and mobile devices increase spatial coverage, though data quality and standardization remain challenges [33].

Multi-Source Data Fusion

- **Techniques:** Statistical and AI-based fusion of radar, satellite, CML, and IoT sensor data improves spatial-temporal rainfall mapping and nowcasting [24] [34] [35] [36].
- **Benefits:** Enhanced accuracy, especially in data-sparse or complex urban terrains.

2.2 Urban Drainage Network Modeling

Integrated 1D-2D Hydrodynamic Modeling

- **Coupled Models:** Simultaneous simulation of sewer (1D), surface (2D), and river networks captures bidirectional interactions and accurately predicts flood extents [15] [37] [38] [39].
- **Software:** Tools like InfoWorks, MIKE-Urban, and SWMM are widely used, with open-source platforms (e.g., UIDS) gaining traction.

Surrogate and Machine Learning Models

- **ANNs and GNNs:** Serve as rapid surrogates for full hydrodynamic simulations, enabling real-time flood prediction with high accuracy and significant computational savings [9] [40] [41].
- **Hybrid Surrogates:** Combine physics-based and data-driven approaches for robust, efficient simulation [11] [42].

High-Resolution DEMs and GIS Tools

- **Role:** High-resolution DEMs and GIS integration improve flow path delineation, drainage area definition, and simulation accuracy [43] [44] [45].
- **Advances:** Deep learning and multi-source data fusion enhance DEM quality and update frequency.

Model Predictive Control (MPC) and Forecast Uncertainty

- **MPC Frameworks:** Optimize drainage operations by incorporating rainfall forecast uncertainty via ensemble, chance-constrained, and tube-based approaches [46] [47] [48].
- **Surrogate Models in MPC:** Replace computationally intensive hydraulic models for real-time control [49].

3. Methods & Data Transparency

3.1 Rainfall Forecasting

- **Data Sources:** Radar, satellite, CML, IoT sensors, video feeds, crowdsourced weather stations.
- **Model Training:** Historical rainfall and flood event data, synthetic data from hydrodynamic simulations, and real-time sensor streams.
- **Validation:** Cross-validation with independent rain gauge networks, satellite flood area analysis, and hydrodynamic model outputs.

3.2 Drainage Network Modeling

- **Network Data:** Sewer maps, DEMs, GIS layers, and urban infrastructure databases.
- **Model Calibration:** Field measurements, remote sensing, and design specifications.
- **Surrogate Model Training:** Outputs from high-fidelity hydrodynamic simulations used to train ANNs, GNNs, and Gaussian Process emulators.

3.3 Integration Frameworks

- **Coupling Strategies:**
  - Direct input of rainfall nowcasts into hydrodynamic or surrogate drainage models.
  - Hybrid frameworks where deep learning models predict both rainfall and drainage responses.
  - GNNs representing drainage networks as graphs, integrating rainfall as node/edge features.
- **Uncertainty Quantification:**
  - Ensemble rainfall forecasts, Bayesian optimization, Monte Carlo simulations, and conformal prediction methods.
- **Computational Efficiency:**
  - Surrogate models, GPU acceleration, and parallel computing for real-time performance.

4. Critical Analysis of Findings

4.1 Hybrid AI and Physics-Based Integration

Hybrid frameworks that combine deep learning rainfall nowcasting with physics-based hydrodynamic models are now state-of-the-art. These systems leverage the rapid inference and pattern recognition of AI with the physical consistency and interpretability of hydrodynamic simulations [1] [2] [3]. For example, LSTM models for rainfall and water level prediction, coupled with CNNs for flood mapping, can mimic the outputs of HEC-RAS 1D/2D models at a fraction of the computational cost, enabling near-real-time urban flood nowcasting [2] [3].

Strengths:

- High accuracy and rapid inference.
- Ability to incorporate heterogeneous data sources.
- Scalability to large urban areas.

Limitations:

- Require large, high-quality training datasets.
- May struggle with generalization to unseen urban morphologies or extreme events.

4.2 Graph Neural Networks for Spatiotemporal Modeling

GNNs have emerged as powerful tools for modeling urban drainage networks, representing manholes as nodes and pipes as edges in directed graphs. Advanced GNN architectures, often combined with attention mechanisms and recurrent units (e.g., GNN-LSTM), capture both spatial dependencies and temporal dynamics, leading to significant improvements in water level and flow predictions [4] [5].

Key Innovations:

- Gated fusion attention mechanisms dynamically balance rainfall inputs and internal network states.
- Physics-constrained GNNs improve interpretability and adherence to hydraulic principles.
- Transferability and scalability demonstrated via fine-tuning and normalization [50].

Challenges:

- Model complexity and interpretability.
- Need for reliable, up-to-date network topology data.

4.3 Multi-Source Data Fusion and Uncertainty Quantification

The fusion of radar, satellite, CML, and IoT sensor data is critical for improving rainfall nowcasting, especially in urban areas with sparse traditional sensors [6] [29]. Ensemble and Bayesian methods are increasingly used to quantify and propagate uncertainties from rainfall forecasts through drainage simulations, supporting probabilistic flood warnings [7] [8].

Benefits:

- Enhanced spatial and temporal resolution of rainfall inputs.
- Improved reliability and robustness of flood nowcasting.

Limitations:

- Data integration and calibration challenges.
- Computational demands for real-time ensemble simulations.

4.4 Surrogate Modeling for Computational Efficiency

Surrogate models, particularly those based on deep neural networks and Gaussian Process emulators, have dramatically reduced the computational burden of urban flood simulations. These models can replicate the outputs of full hydrodynamic simulations with high accuracy and speedups of 10–1000x, enabling real-time flood prediction and control [9] [10] [11].

Strengths:

- Real-time applicability for early warning and operational control.
- Flexibility to integrate with AI-based rainfall nowcasting.

Limitations:

- Potential loss of physical interpretability.
- Challenges in modeling complex 1D drainage components [9].

4.5 Explainability and Model Transparency

Explainable AI (XAI) methods, such as SHAP and LIME, are increasingly integrated into hybrid nowcasting frameworks to provide human-understandable explanations of model predictions. This enhances trust and supports decision-making in urban flood management [5] [12] [13].

Opportunities:

- Identification of key flood drivers (e.g., rainfall intensity, imperviousness).
- Improved stakeholder confidence and actionable insights.

Remaining Issues:

- Need for standardized evaluation metrics for explainability.
- Human-centered design to ensure usability in operational contexts.

5. Real-World Implications

Operational Urban Flood Nowcasting

- **Early Warning:** Integrated frameworks enable timely, location-specific flood warnings, supporting emergency response and infrastructure management [3] [51].
- **Real-Time Control:** MPC and surrogate models facilitate dynamic drainage system operations, reducing flood risk during extreme rainfall [47] [48].
- **Resource Optimization:** Data fusion and AI-driven predictions support efficient allocation of flood mitigation resources (e.g., pump activation, road closures).

Urban Planning and Resilience

- **Infrastructure Design:** High-resolution, data-driven models inform the design and retrofitting of urban drainage systems for future climate scenarios [43] [44].
- **Risk Assessment:** Probabilistic and impact-based nowcasting frameworks enable targeted flood risk zoning and insurance planning [7] [52].

Societal and Policy Impact

- **Transparency and Trust:** XAI and explainable models foster public trust and facilitate communication between scientists, policymakers, and the public [5] [12].
- **Inclusivity:** Crowdsourced and human-sensed data streams democratize flood monitoring and enhance situational awareness in underserved communities [33] [53].

6. Future Research Directions

6.1 Systematic Evaluation and Benchmarking

- **Multi-City Studies:** Conduct large-scale, cross-city evaluations of integrated nowcasting frameworks under diverse urban morphologies and meteorological regimes [54].
- **Standardized Benchmarks:** Develop open datasets and benchmarking platforms for fair comparison of AI, surrogate, and hybrid models [9].

6.2 Uncertainty Quantification and Propagation

- **Advanced Methods:** Refine ensemble, Bayesian, and conformal prediction techniques for robust uncertainty quantification from rainfall input to flood impact [55] [56].
- **Operational Integration:** Develop real-time, uncertainty-aware decision support systems for urban flood management [7] [8].

6.3 Adaptive and Explainable Hybrid Frameworks

- **Continual and Reinforcement Learning:** Explore adaptive systems that learn from real-time sensor feedback and evolving urban landscapes [57] [58].
- **Explainable AI:** Advance XAI methods tailored to urban flood nowcasting, ensuring actionable and trustworthy outputs for practitioners [5] [12].

6.4 Integration of Human-Sensed and Physical Data

- **Crowdsourcing and IoT:** Expand the use of human-sensed data (e.g., social media, mobile apps) in conjunction with physical sensors for comprehensive flood monitoring [33] [53].
- **Data Fusion Algorithms:** Develop scalable, robust algorithms for integrating heterogeneous data streams in real time [29] [36].

Synthesis

The field of urban flood nowcasting is rapidly advancing through the integration of high-resolution, multi-source rainfall forecasts with sophisticated urban drainage network models. Hybrid AI-physics frameworks, graph neural networks, and surrogate modeling have collectively improved prediction accuracy, computational efficiency, and operational applicability. Data fusion and uncertainty quantification are central to these advances, while explainable AI is increasingly recognized as essential for transparency and trust.

Despite these achievements, challenges remain in standardizing uncertainty quantification, ensuring model generalizability, and scaling integrated systems to diverse urban environments. Future research should prioritize systematic evaluation, adaptive and explainable frameworks, and the integration of human-sensed data to further enhance the reliability and societal impact of urban flood nowcasting systems [1] [2] [4] [5] [6] [7] [8].

References

1. Hybrid AI-IoT System for Real-Time Flood Prediction and Pump Automation in Urban Areas Khuluq, M.K., Taufiqurrahman, Gunawan, A.I. 2025 International Electronics Symposium, IES 2025, 2025 https://www.scopus.com/pages/publications/105018052476?origin=scopusAI

2. Enhancing flood forecasting with deep learning: A scalable alternative to traditional hydrodynamic models Duangkhwan, W., Ekkawatpanit, C., Petpongpan, C., (...), Jaturapitakkul, C. Environmental Modelling and Software, 2026 https://www.scopus.com/pages/publications/105026861048?origin=scopusAI

3. Application of Deep Learning and Rain-On-Grid Hydrodynamic Modeling for Catchment-Scale Flood Forecasting in Guwahati, India Ashok, S.P., Pekkat, S. Journal of Flood Risk Management, 2026 https://www.scopus.com/pages/publications/105040250742?origin=scopusAI

4. Multi-Source Data Fusion and Intelligent Flood Risk Prediction for Urban Drainage Networks Based on Graph Neural Networks Zou, X. 2025 4th International Conference on Cloud Computing, Big Data Application and Software Engineering, CBASE 2025, 2025 https://www.scopus.com/pages/publications/105033495473?origin=scopusAI

5. Explainable Urban Flood Prediction by Explainable Spatio-Temporal Graph Neural Network with Interpretable Risk Alerts Sathiyamoorthy, M. 6th International Conference on Inventive Computation and Information Technologies, ICICIT 2026 - Proceedings, 2026 https://www.scopus.com/pages/publications/105047934629?origin=scopusAI

6. On the potential of commercial microwave link networks for high spatial resolution rainfall monitoring in urban areas Zheng, X., Messer, H., Wang, Q., (...), Yang, T. Atmospheric Research, 2022 https://www.scopus.com/pages/publications/85133409341?origin=scopusAI

7. Bayes_Opt-SWMM: A Gaussian process-based Bayesian optimization tool for real-time flood modeling with SWMM Tanim, A.H., Smith-Lewis, C., Downey, A.R.J., (...), Goharian, E. Environmental Modelling and Software, 2024 https://www.scopus.com/pages/publications/85196630789?origin=scopusAI

8. Nowcasting of rainfall and of combined sewage flow in urban drainage systems Achleitner, S., Fach, S., Einfalt, T., Rauch, W. Water Science and Technology, 2009 https://www.scopus.com/pages/publications/65649135880?origin=scopusAI

9. UrbanFloodBench: Bridging AI and hydrology through benchmarking of coupled 1D–2D urban flood surrogate models Lim, J.Y., Herath, H.M.V.V., Rasnayaka, S., (...), Cheshenko, D. Environmental Modelling and Software, 2026 https://www.scopus.com/pages/publications/105051164726?origin=scopusAI

10. A hybrid framework for real-time flash flood forecasting in small ungauged catchments: integrating hydrodynamic simulations with LSTM networks An, H., Ouyang, C. Journal of Hydrology, 2025 https://www.scopus.com/pages/publications/105008353040?origin=scopusAI

11. Accelerating hydrodynamic simulations of urban drainage systems with physics-guided machine learning Palmitessa, R., Grum, M., Engsig-Karup, A.P., Löwe, R. Water Research, 2022 https://www.scopus.com/pages/publications/85136018118?origin=scopusAI

12. An Explainable Artificial Intelligence (XAI) Framework for Deep Learning Based Classification to Generate Textual Explanations on Predicted Images Sheela, B.P., Girisha, H. International Journal of Intelligent Engineering and Systems, 2024 https://www.scopus.com/pages/publications/85207950180?origin=scopusAI

13. An Explainable AI Framework for Flood Severity Prediction Using SHAP and LIME Interpretability Patil, D., Patil, B., Kulkarni, A., (...), Thorat, N. 2026 International Conference on Intelligent and Sustainable Electronics and Computing Technologies, INSECT 2026, 2026 https://www.scopus.com/pages/publications/105051469682?origin=scopusAI

14. Remote sensing based rain fall prediction using big data assisted integrated routing framework Refonaa, J., Lakshmi, M. Journal of Ambient Intelligence and Humanized Computing, 2021 https://www.scopus.com/pages/publications/85098778828?origin=scopusAI

15. An integrated modeling approach to predict flooding on urban basin Dey, A.K., Kamioka, S. Water Science and Technology, 2007 https://www.scopus.com/pages/publications/33947641138?origin=scopusAI

16. A Comprehensive review of Short-Term Weather Forecasting Using Artificial Intelligence Techniques Based on Internet of Things Setiowati, S., Al Bukhori, J., Hartono, T.T., (...), Nuralam Proceedings - IC2IE 2025: 8th 2025 International Conference of Computer and Informatics Engineering: Human-Machine Synergy Brings Together the Physical and Digital Worlds, 2025 https://www.scopus.com/pages/publications/105031763785?origin=scopusAI

17. A deep learning model coupled with metaheuristic optimization for urban rainfall prediction Zhao, W., Zhang, Z., Khodadadi, N., Wang, L. Journal of Hydrology, 2025 https://www.scopus.com/pages/publications/85213514941?origin=scopusAI

18. A VMD–PSO hybrid framework with machine and deep learning for Aaccurate spatial–temporal daily rainfall occurrence prediction in Pakistan’s diverse climatic regions Farman, H., Hussain, M.A., Shaikh, S., Hassan, S. Modeling Earth Systems and Environment, 2026 https://www.scopus.com/pages/publications/105039490486?origin=scopusAI

19. Hybrid artificial intelligence models based on adaptive neuro fuzzy inference system and metaheuristic optimization algorithms for prediction of daily rainfall Pham, B.T., Bui, K.-T.T., Prakash, I., Ly, H.-B. Physics and Chemistry of the Earth, 2024 https://www.scopus.com/pages/publications/85186767787?origin=scopusAI

20. Enhancing daily rainfall prediction in urban areas: a comparative study of hybrid artificial intelligence models with optimization algorithms Sheikhi, Y., Ashrafi, S.M., Nikoo, M.R., Haghighi, A. Applied Water Science, 2023 https://www.scopus.com/pages/publications/85176043668?origin=scopusAI

21. A Comparison of Machine Learning Models for Predicting Rainfall in Urban Metropolitan Cities Kumar, V., Kedam, N., Sharma, K.V., (...), Alluqmani, A.E. Sustainability (Switzerland), 2023 https://www.scopus.com/pages/publications/85172891363?origin=scopusAI

22. A novel AI-driven framework for trusted decisions and sustainable flood management in smart cities Shaikh, F., Devare, M. African Journal of Science, Technology, Innovation and Development, 2026 https://www.scopus.com/pages/publications/105047413903?origin=scopusAI

23. Retrieval of rainfall fields in urban areas using attenuation measurements from commercial microwave links: A feasibility study Zohidov, B., Andrieu, H., Servières, M., Normand, N. Rainfall in Urban and Natural Systems - Proceedings of the 10th International Workshop on Precipitation in Urban Areas, UrbanRain 2015, 2015 https://www.scopus.com/pages/publications/85070946224?origin=scopusAI

24. New algorithm for integration between wireless microwave sensor network and radar for improved rainfall measurement and mapping Liberman, Y., Samuels, R., Alpert, P., Messer, H. Atmospheric Measurement Techniques, 2014 https://www.scopus.com/pages/publications/84908101843?origin=scopusAI

25. Use of Radar Quantitative Precipitation Estimates (QPEs) for Improved Hydrological Model Calibration and Flood Forecasting Wijayarathne, D., Coulibaly, P., Boodoo, S., Sills, D. Journal of Hydrometeorology, 2021 https://www.scopus.com/pages/publications/85118499227?origin=scopusAI

26. Comparing GPROF V7 and commercial microwave link precipitation estimates with DPR in the Netherlands, Sri Lanka and Nigeria Bogerd, L., Walraven, B., Overeem, A., (...), Uijlenhoet, R. Hydrological Sciences Journal, 2026 https://www.scopus.com/pages/publications/105045154550?origin=scopusAI

27. Calibration of Weather Radar Rainfall Field Based on Microwave Links Zhang, P., Liu, X., Zhou, Z., (...), Yang, P. Meteorological Monthly, 2023 https://www.scopus.com/pages/publications/85187404605?origin=scopusAI

28. Integrated Rainfall Estimation Using Rain Gauges and Weather Radar: Implications for Rainfall-Induced Landslides De Biase, M., Lupiano, V., Chiaravalloti, F., (...), Pisano, L. Remote Sensing, 2025 https://www.scopus.com/pages/publications/105021459052?origin=scopusAI

29. Spatiotemporal Accuracy Assessment of Multi-source Fusion Precipitation Product SUPER in the HanJiang River Basin Liu, S., Wei, L., Dong, J., (...), Qi, D. Journal of Geo-Information Science, 2024 https://www.scopus.com/pages/publications/85193724523?origin=scopusAI

30. A review of video-based rainfall measurement methods Yan, K., Chen, H., Hu, L., (...), Guo, S. Wiley Interdisciplinary Reviews: Water, 2023 https://www.scopus.com/pages/publications/85162691079?origin=scopusAI

31. A dual-channel framework for video-based rainfall intensity estimation using weakly supervised rain streak extraction Xu, J., Guo, N., Sun, F., Wu, F. Journal of Hydrology, 2026 https://www.scopus.com/pages/publications/105051073784?origin=scopusAI

32. Toward accurate and scalable rainfall estimation using surveillance camera data and a hybrid deep-learning framework Manuel, F.-S., Khu, S.-T., Guan, J., Wang, M. Environmental Science and Ecotechnology , 2025 https://www.scopus.com/pages/publications/105003840462?origin=scopusAI

33. Gauging Through the Crowd: A Crowd-Sourcing Approach to Urban Rainfall Measurement and Storm Water Modeling Implications Yang, P., Ng, T.L. Water Resources Research, 2017 https://www.scopus.com/pages/publications/85038922655?origin=scopusAI

34. High-Resolution Simulation Study Exploring the Potential of Radars, Crowdsourced Personal Weather Stations, and Commercial Microwave Links to Monitor Small-Scale Urban Rainfall de Vos, L.W., Raupach, T.H., Leijnse, H., (...), Uijlenhoet, R. Water Resources Research, 2018 https://www.scopus.com/pages/publications/85059067873?origin=scopusAI

35. A Statistical Approach to Retrieve Rainfall Intensity by Attenuation Measurements From Commercial Microwave Links Zheng, X., Messer, H., Qin, Y., (...), Yang, T. Earth's Future, 2023 https://www.scopus.com/pages/publications/85176015833?origin=scopusAI

36. Rainfall estimation from a German-wide commercial microwave link network: Optimized processing and validation for 1 year of data Graf, M., Chwala, C., Polz, J., Kunstmann, H. Hydrology and Earth System Sciences, 2020 https://www.scopus.com/pages/publications/85086766567?origin=scopusAI

37. A multi-level and modular model for simulating the urban flooding and its application to Tianjin City Li, D., Wang, X., Xie, Y., (...), Cai, X. Natural Hazards, 2016 https://www.scopus.com/pages/publications/84961141803?origin=scopusAI

38. UIDS: A Matlab-based urban flood model considering rainfall-induced and surcharge-induced inundations Tran, V.N., Kim, J. Environmental Modelling and Software, 2024 https://www.scopus.com/pages/publications/85197386470?origin=scopusAI

39. Advanced 2-D modeling for flood reduction studies and storm drain master planning Diaz, C., Jorgensen, R. WEFTEC 2012 - 85th Annual Technical Exhibition and Conference, 2012 https://www.scopus.com/pages/publications/85070477023?origin=scopusAI

40. RAPIDS: Early warning system for urban flooding and water quality hazards Duncan, A.P., Chen, A.S., Keedwell, E.C., (...), Savić, D.A. Machine Learning in Water Systems - AISB Convention 2013, 2013 https://www.scopus.com/pages/publications/84894223470?origin=scopusAI

41. Graph neural network-based surrogate modelling for real-time hydraulic prediction of urban drainage networks Zhang, Z., Tian, W., Lu, C., (...), Yuan, Z. Water Research, 2024 https://www.scopus.com/pages/publications/85200113651?origin=scopusAI

42. Toward real-time high-resolution fluvial flood forecasting: A robust surrogate approach based on overland flow models Nguyen, G.V., Van, C.P., Tran, V.N., (...), Lee, G. Environmental Modelling and Software, 2026 https://www.scopus.com/pages/publications/105018920255?origin=scopusAI

43. The influence of digital elevation model resolution on overland flow networks for modelling urban pluvial flooding Leitão, J.P., Boonya-aroonnet, S., Prodanović, D., Maksimović, Č. Water Science and Technology, 2009 https://www.scopus.com/pages/publications/77649230116?origin=scopusAI

44. An urban DEM reconstruction method based on multisource data fusion for urban pluvial flooding simulation Huang, H., Liao, W., Lei, X., (...), Wang, H. Journal of Hydrology, 2023 https://www.scopus.com/pages/publications/85144817186?origin=scopusAI

45. Relating DEM spatial resolution and hyetograph temporal resolution to flood modeling accuracy Abedin, S.J.H., Stephen, H. World Environmental and Water Resources Congress 2015: Floods, Droughts, and Ecosystems - Proceedings of the 2015 World Environmental and Water Resources Congress, 2015 https://www.scopus.com/pages/publications/84935039214?origin=scopusAI

46. Model Predictive Control of Urban Drainage Systems Considering Uncertainty Svensen, J.L., Sun, C., Cembrano, G., Puig, V. IEEE Transactions on Control Systems Technology, 2023 https://www.scopus.com/pages/publications/85164391391?origin=scopusAI

47. Predictive MPC-Based Operation of Urban Drainage Systems Using Input Data-Clustered Artificial Neural Networks Rainfall Forecasting Models Jafari, F., Mousavi, S.J., Ponnambalam, K. Hydrology, 2023 https://www.scopus.com/pages/publications/85166530283?origin=scopusAI

48. Sewer orientated framework for ensemble-based chance-constrained model predictive control Lorenz Svensen, J., Henrik Niemann, H., Falk, A.K.V., Poulsen, N.K. Advanced Control for Applications: Engineering and Industrial Systems, 2021 https://www.scopus.com/pages/publications/85136027585?origin=scopusAI

49. Real-time model predictive control of urban drainage system in coastal areas Chen, Y., Wang, C., Huang, H., (...), Wang, Z. Journal of Hydrology, 2024 https://www.scopus.com/pages/publications/85179853398?origin=scopusAI

50. Evaluation of graph neural networks for urban drainage metamodeling: Key components and transferability analysis Garzón, A., Kapelan, Z., Langeveld, J., Taormina, R. Water Research, 2026 https://www.scopus.com/pages/publications/105024411715?origin=scopusAI

51. Early Warning System of Urban Waterlogging under the Threat of Extreme Rainfall Events Cao, X., Qi, Y., Li, M., (...), Ni, G. Chinese Journal of Atmospheric Sciences, 2022 https://www.scopus.com/pages/publications/85137035644?origin=scopusAI

52. Enhancing flood risk assessment in urban areas by integrating hydrodynamic models and machine learning techniques Khoshkonesh, A., Nazari, R., Nikoo, M.R., Karimi, M. Science of the Total Environment, 2024 https://www.scopus.com/pages/publications/85202963432?origin=scopusAI

53. A spatial–temporal graph deep learning model for urban flood nowcasting leveraging heterogeneous community features Farahmand, H., Xu, Y., Mostafavi, A. Scientific Reports, 2023 https://www.scopus.com/pages/publications/85154538932?origin=scopusAI

54. A two-phase physics-informed hybrid deep learning model for flood forecasting: improving process learning and interpretability Chen, C., Li, B., Zhao, Y., (...), Liang, Z. Advances in Water Resources, 2026 https://www.scopus.com/pages/publications/105037799059?origin=scopusAI

55. Uncertainty Quantification for Flood Forecasting in Small Catchments Spils, M., Tomforde, S. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105023827513?origin=scopusAI

56. Estimating uncertainty in flood model outputs using machine learning informed by Monte Carlo analysis Nguyen, M., Wilson, M.D., Lane, E.M., (...), Pearson, R.A. Journal of Hydrology, 2025 https://www.scopus.com/pages/publications/105015138168?origin=scopusAI

57. A Smart Water Distribution Network Management Framework Based on Internet of Things and Reinforcement Learning Technologies Aidan, I.A., Aljumaily, M.M., Almaqbali, H.A.M., Hussein, A.H. Journal of Soft Computing and Data Mining, 2026 https://www.scopus.com/pages/publications/105043713276?origin=scopusAI

58. Real-Time Dynamic Urban Flood Prediction: An Integrated Deep Learning Architecture with Adaptive Updating Wang, S., Xiang, Z., Song, Y., (...), Tao, T. ACS ES and T Water, 2026 https://www.scopus.com/pages/publications/105041612707?origin=scopusAI